

HTT-BRANDS Control-Tower Design System

1) Overview & Principles

Purpose

- Provide a unified, scalable UI for the Control-Tower, enabling rapid, consistent delivery.

Goals

- Consistency across modules
- Efficiency via reusable tokens/components
- Scalability for new features

Audience

- Designers, developers, product managers

Design Principles

- Data-dense, clarity-first, operationally safe
- Prefer progressive disclosure, auditability, and low cognitive load
- Accessibility: WCAG 2.1 AA as a baseline
- Responsive: Mobile-friendly; optimized for desktop $\geq 1280\text{px}$; adaptive at 1440, 1200, 992, 768, 480
- Performance: Ship only needed assets; virtualize large lists; lazy-load non-critical modules; target $< 200\text{ms}$ perceived interactions

Practical Guidance

- Lead with the most critical signals; defer details with drill-in and tooltips
- Make risky actions explicit and reversible (confirmations, undo)
- Favor inline editing for speed; use modals for complex or destructive flows
- Instrument usage; use telemetry to validate additions and inform deprecations

2) Foundation

2.1 Colors

Palette

- Primary: HTT Blue #0046FF; Dark #0836B8; Light #E6EFFF
 - Secondary: Teal #00B3A4; Amber Accent #FFB000
 - Neutrals: N0 #FFFFFF, N50 #F5F7FA, N100 #E6EAF0, N300 #A8B0BD, N600 #5A6472, N900 #1E242B
 - Semantic: Success #1A7F37, Warning #B54708, Error #B42318, Info #0B6BBD
- #### Usage
- Prefer light tints for background states; ensure $\geq 7:1$ contrast for overlaid text where possible
 - Maintain contrast: $\geq 4.5:1$ for text; $\geq 3:1$ for large text/graphics
 - Reserve Amber for attention and accents; avoid overuse to prevent alarm fatigue

2.2 Typography

Families

- Inter, Segoe UI, Helvetica, Arial, sans-serif
- Scale (font size/line height)
- H1 32/40, H2 24/32, H3 20/28, H4 18/26, Body 16/24, Small 14/20, Caption 12/16
- Weights
- 600 for headings, 500 for buttons, 400 for body
- Avoid 700 except dashboard titles
- Responsive
- Reduce one step under 768px (e.g., H2 → H3)
- Usage
- Keep headings concise; one H1 per view
- Favor sentence case; avoid all-caps for readability except for tags/badges if needed

2.3 Spacing

System

- Base 8px
- Scale: 4, 8, 16, 24, 32, 48, 64, 96
- Tokens: s-1...s-8 (map to the scale above)
- Rules
- Margins outside, padding inside
- Use spacing tokens exclusively to maintain rhythm

2.4 Elevation

States

- Hover: subtle elevation/color change
- Focus: 2px visible ring (see Accessibility)
- Active: 95% scale or pressed color
- Usage
- Use elevation sparingly to indicate interactivity and layering
- Keep overlays (menus, popovers, dialogs) clearly above base content without excessive shadow

2.5 Iconography

Sizes

- 16/20/24/32px
- Usage
- Use icons to reinforce labels; do not replace critical text
- Align to pixel grid; maintain consistent optical weight
- Prefer outlined or filled styles consistently per context

3) Layout & Grid System

Grid

- 12 columns
- Minimum content width: 72px
- Gutters: 24px
- Page padding: 32px
- Responsive
- Optimized desktop ≥ 1280 px
- Adaptive breakpoints at 1440, 1200, 992, 768, 480
- Collapse columns progressively; maintain touch targets $\geq 44 \times 44$ px on smaller viewports
- Composition
- Use the grid for alignment of all structural elements (nav, content, side panels)
- Keep primary content in the first visible columns; place utilities/filters in side regions or top toolbars
- Maintain consistent vertical rhythm using spacing tokens
- Page Templates
- Dashboard: KPI strip + widgets grid
- List/Table: Filters, toolbar, table, drawer
- Detail: Header, tabs, activity
- Settings: Left nav + forms
- Profile: Account, preferences, API keys
- Guidance
- Start with the target template, then compose with organisms and molecules
- Reserve left navigation for persistent sections; use drawers for contextual details/actions

4) Component Library with Usage Guidelines

Atoms

- Buttons (Primary, Secondary, Tertiary, Ghost; Loading; Disabled)
 - Use Primary for the main action per view; Secondary for alternatives; Tertiary/Ghost for low-emphasis or inline actions
 - Show Loading on async; set Disabled to prevent invalid submissions
- Inputs (types + success/error)
 - Provide clear labels, helper text, and error messages; indicate success/error states inline
- Checkbox/Radio
 - Use Checkbox for multi-select; Radio for mutually exclusive choices
- Switch
 - Use for immediate, reversible binary toggles
- Icons (16/20/24/32px)
 - Pair with labels for clarity when space allows
- Badge/Tag (status colors)
 - Reflect semantic states; keep labels short
- Avatar (24–40px)
 - Use for users/entities; fall back to initials when images are absent
- Molecules
- Form Group (label, helper, error)
 - Ensure accessible labeling: group related inputs

Ensure accessible loading, group related inputs

- Search Bar (debounce 300ms)
 - Debounce input; show inline loading; preserve query on navigation
- Dropdown (typeahead, max-height 320px)
 - Support keyboard navigation; truncate long options with tooltip
- Tooltip/Popover (delay 150ms)
 - Use Tooltip for hints; Popover for rich, actionable content
- Breadcrumbs
 - Reflect current location; last item non-interactive
- Pagination (compact and verbose)
 - Choose compact for dense tables; verbose when clarity is needed
- Alert Banner (inline/global)
 - Inline for local context; global for system-wide states
- Organisms
- Nav
 - Top: include environment context; Side: section navigation with clear hierarchy
- Data Table
 - Features: sticky header, sort, filter, bulk select, row actions
 - Virtualize large lists; preserve column alignment; provide empty/loaded/error states
- Cards (KPIs, drill-in)
 - Keep KPIs concise; support drill-in for detail
- Modals/Dialogs (focus trap)
 - Use for blocking flows; trap focus; restore focus on close
- Forms (multi-step with progress)
 - Show progress; support optimistic UI where safe; validate inline
- Widgets (live status, SLAs, queues)
 - Update asynchronously; avoid noisy auto-refresh; provide timestamps
- Patterns
- Auth: email+SSO, 2FA
- Data entry: inline and modal with optimistic UI
- Empty states: icon, message, primary action
- Loading: skeletons
- Errors: retry + support link
- Confirmations: destructive actions require typed confirm
- Bulk actions: summary + undo

5) Interaction Patterns

Animation

- Durations: Fast 120–180ms, Medium 220–320ms, Slow 420–520ms
- Easing: cubic-bezier(0.2, 0, 0, 1)
- Micro-states
- Hover: subtle elevation/color shift
- Focus: 2px visible ring

• .focus-visible { outline: 2px solid #000; }

- Active: 95% scale or pressed color
Behavior
- Use progressive disclosure for complex workflows (tabs, accordions, drawers)
- Preserve user context (filters, sort, pagination) across navigations
- Provide auditability (surface who/when for critical changes)

6) Accessibility Requirements

Keyboard

- Logical tab order; Esc closes dialogs and menus
- Arrow keys in menus/grids; Enter/Space activate
- Restore focus on close to the triggering element

Screen Readers

- Provide ARIA roles and labels
- Use live regions for async updates

Focus

- Visible ring color #0B6BBD
- Ring thickness 2px; ensure contrast against backgrounds

Touch

- Targets $\geq 44 \times 44$ px
- Ensure sufficient spacing to avoid accidental taps

Contrast

- Text $\geq 4.5:1$; large text/graphics $\geq 3:1$

7) Implementation Standards

API & Naming

- Props: clear names (size, variant, state, icon, onAction)
- Controlled components by default
- Tokens: kebab-case; Components: PascalCase
Code
- TypeScript
- Unit tests and a11y tests required
- Story-based docs with examples and edge cases

Performance

- Ship only needed assets
- Virtualize large lists
- Lazy-load non-critical modules
- Aim for <200ms perceived interactions

Versioning

- SemVer
- Changelog for every release
- Deprecations include migration notes

8) Governance

Roles

- Design owner, Code owner, Accessibility lead, Docs maintainer
Contribution Workflow
- RFC → design review → a11y check → code PR → release
Cadence
- Reviews biweekly
- Maintenance monthly audits
- Telemetry guides future additions



